Comparative Review of Detection Methods

From manual propaganda to adaptive influence systems

- Information warfare manipulates perception, trust, and behaviour
- ML scales generation, targeting, and amplification
- Social media is now an operational battlespace
- Actors include states, platforms, defence bodies, and researchers

1. Singer, P.W. and Brooking, E.T. (2018) *LikeWar: The Weaponization of Social Media*. Boston, MA: Houghton Mifflin Harcourt.

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Detection Target	ML Method	**Input features**	Main strength	Main limitation
Machine / Bot generated text	CNN + FastText embeddings	Word patterns, semantic vectors	Strong benchmark detection of generated tweets	Text methods are fast and accurate on known datasets, but attackers can adapt language models to reduce performance.
Coordinated behaviour	Graph-based models, inverse RL	Retweet networks, temporal activity, account interactions	Retweet networks, temporal activity, account interactions	Graph methods are often more robust because they detect coordination patterns rather than only the text itself
Deepfakes / Synthetic content	Deep learning + embeddings	Frequency artefacts, visual inconsistencies	Learns machine-generated patterns	Synthetic media detection is an arms race because generative models evolve faster than static forensic cues
Contested environments. Real-time threat triage.	Robust detection framing. Random Forest + TensorFlow pipelines.	Text, metadata, network signals	Highlights operational realism	Adversaries adapt to detectors. Operational systems work best as triage tools for analysts rather than fully autonomous decision-makers.

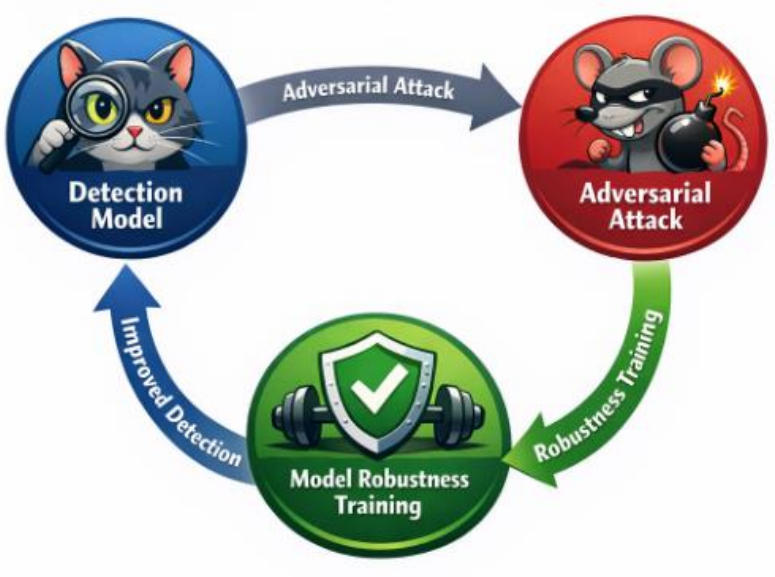
2. Dabholkar, S. et al. (2024) 'Adversarial attacks on reinforcement learning agents for command and control'. Add final journal, volume, issue and page details from publisher record before submission.
3. Duddu, V. (2018) 'A Survey of Adversarial Machine Learning in Cyber Warfare', *Defence Science Journal*, 68(4), pp. 356–366. doi: 10.14429/dsj.68.12731.
4. Sadiq, S., Aljrees, T. and Ullah, S. (2023) 'Deepfake Detection on Social Media: Leveraging Deep Learning and FastText Embeddings for Identifying Machine-Generated Tweets', *IEEE Access*. doi: 10.1109/ACCESS.2023.3308515.

Robustness, Explainability, and Research Direction

The real challenge is adversarial adaptation

- Adversarial ML exposes poisoning and evasion weaknesses
- High accuracy does not guarantee operational reliability
- RL systems are vulnerable to manipulated inputs
- Future systems must be robust, explainable, and layered

The Cat and Mouse loop



1. Singer, P.W. and Brooking, E.T. (2018) *LikeWar: The Weaponization of Social Media*. Boston, MA: Houghton Mifflin Harcourt.